

3D Modeling and Control of Fuel Sloshing in A Spacecraft

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Abstract— Fuel sloshing is a critical problem in attitude control of spacerobots. This paper presents a 3D mathematical model to describe dynamic of fuel slosh in spacerobot's container. A multi-pendulum model is considered for characterization of the most prominent sloshing modes. After modeling, a nonlinear feedback control law is designed to control the attitude and translation velocity vectors of the spacerobot in orbital maneuvering. In this paper, spacerobot is modeled by a rigid body and the whole fuel is modeled by a fixed mass and two sloshing masses. Also, the control subjects are three torques about the rigid body and two gimbal deflection angles.

Keywords—fuel sloshing; multi-pendulum model; 3D modeling; Lyapunov control

I. INTRODUCTION

The space programs are cynosure, some of this programs in new aspects as spacecraft formation [1-3], spacecraft rendezvous [4], space objects collision [5], and site location in LEO missions [6-9] are knowledge-based innovation. Space orbital maneuver [10,11] and spacecraft attitude control [12-14] are based on dynamic modeling [15-17]. Design of the control systems is essential for the other systems such as attitude of quadrotor [18,19]. The high percent of mass of the large spacerobots is allocated to fuel, and this fuel uses as a thrust to orbital maneuvering. If axial or angular acceleration of the spacecraft changes, the fuel in the container will be affected with sloshing [20]. This harmful problem could fail the spacerobot's mission. So, the fuel sloshing is one of the most important problems in attitude control. Interaction between fuel sloshing dynamics and attitude dynamics is well known object in literature of attitude control. The previous works presented nonlinear modeling and control based on Lyapunov theory for a planar maneuvering. Some of these studies used single pendulum model [21,22] or single mass-spring model [22] for characterization the lowest frequency of sloshing mode, and the other studies used multi-pendulum model [23] or multi-mass-spring model [24,25] for characterization the most prominent sloshing modes.

In the previous works multi-pendulum models were in 2D space. So, in this paper, a multi-pendulum model in 3D space is presented to characterize the most prominent sloshing modes.

In the second section of this paper a spacerobot with a partially filled spherical fuel container that maneuver in 3D space is considered, dynamics of fuel sloshing is modeled by a 3D multi-pendulum model, and coupled equations of the spacerobot and fuel motion is derived. In the third section after presenting candidate Lyapunov function, a nonlinear controller based on candidate function is developed, the objectives of this control system are stabilization of attitude and translation velocity vectors of the spacerobot, as seen in Fig.1, the subjects of this control system are three torques (M_x , M_y and M_z) about the rigid body and two gimbal deflection angles (δ_2, δ_1). In the fourth section a physical example is simulated to describe effectiveness of designed controller on the nonlinear multi-pendulum system.

II. MODELING

In this section, equations of coupled dynamics of the spacerobot and fuel sloshing is derived. The total mass of the spacecraft is modeled by a rigid body (m) with moment of inertia (I), that it is fixed on the center of mass of the spacerobot. The fuel container is spherical and fuel is modeled by a rigidly attached mass (m_0) with moment of inertia (I_0) and sloshing masses that demonstrated by pendulum masses (m_i , $i = 1, 2, \dots, N$) with moment of inertia (I_i).

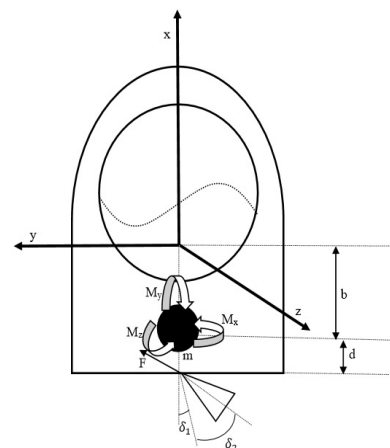


Fig. 1. A spacecraft with a fuel container

The distances between center of container, location of m_0 and location of m_i are h_0 and h_i , respectively. Also, the distances between center of mass of the spacecraft, pivot gimbal and center of container are d and b , respectively. Geometric characteristics of container, kind of fuel, and fill ratio of fuel container are effective factors on parameters of the multi-pendulum model ($m_0, I_0, h_0, m_i, I_i, h_i$). The coupled equations of the spacecraft and fuel motion can derive from Kirochoff's equations [26] and extension of mechanic Lagrangian to include non-conservative forces:

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{V}} + \bar{\Omega} \times \frac{\partial L}{\partial \dot{V}} = \tau_t \quad (1)$$

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{\Omega}} + \bar{\Omega} \times \frac{\partial L}{\partial \dot{\Omega}} + \bar{V} \times \frac{\partial L}{\partial \dot{V}} = \tau_r \quad (2)$$

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{\zeta}} - \frac{\partial L}{\partial \zeta} + \frac{\partial R}{\partial \dot{\zeta}} = 0 \quad (3)$$

Where $V = [v_x \ v_y \ v_z]^T$, $\Omega = [\Omega_1 \ \Omega_2 \ \Omega_3]^T$ are the vector of axial and vector of angular velocity of the rigid body (m), respectively. $\zeta = [\varphi \ \theta]$ is the vector of pendulum masses coordinates, and τ_t, τ_r are the vectors of generalized forces and moments that act on center of mass of the spacerobot, respectively. In above equations for a given vector p , $\bar{p}$ is skew-symmetric matrix, and Lagrangian is expressed in terms of $V, \Omega, \zeta, \dot{\zeta}$. Also R is Rayleigh dissipation function:

$$R = \sum_{i=1}^N \varepsilon_i (\dot{\varphi}_i^2 + \dot{\theta}_i^2) \quad (4)$$

Where, ε_i is damping coefficient of i -th mass. For derive vectors of axial velocity, at first according to Fig. 1 and Fig. 2 position vectors of rigid body (m), and fuel masses (m_0, m_i) can be written, respectively:

$$r_c = [x - b \ y \ z]^T, r_0 = [x - h_0 \ y \ z]^T,$$

$$r_i = [x + h_i - l_i \cos \varphi_i \ y + l_i \sin \varphi_i \sin \theta_i \ z + l_i \sin \varphi_i \cos \theta_i]$$

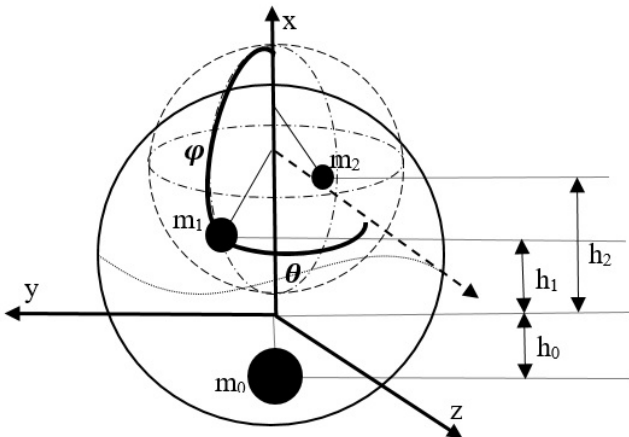


Fig. 2. A 3D multi-pendulum model of fuel sloshing

Where l_i is length of i -th pendulum. Then velocity vectors can derive:

$$V_c = \dot{r}_c + \Omega \times r_c, V_0 = \dot{r}_0 + \Omega \times r_0, V_i = \dot{r}_i + \Omega \times r_i, \\ V = \begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{z} \end{bmatrix} + \Omega \times \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

Where V_c is vector of axial velocity of rigid body, V_0, V_i are vectors of axial velocity of fuel masses, and V is vector of axial velocity of center of un-turbulent fuel. Lagrangian is equal to total kinetic energy minus potential energy ($L = T - U$), and because we assume that the spacerobot is in zero gravity environment, and pendulums have no potential energy, Thus Lagrangian is equal to total kinetic energy, so Lagrangian be written as:

$$L = \frac{1}{2} m V_c^2 + \frac{1}{2} m_0 V_0^2 + \frac{1}{2} \Omega^T (J + I_0 I) \Omega + \frac{1}{2} \sum_{i=1}^N [m_i V_i^2 + (\omega_i + \Omega)^T (I_i I) (\omega_i + \Omega)] \quad (5)$$

where

$$\omega_i = \begin{bmatrix} \dot{\varphi}_i \\ \dot{\theta}_i \end{bmatrix}, J = \begin{bmatrix} j_x & 0 & 0 \\ 0 & j_y & 0 \\ 0 & 0 & j_z \end{bmatrix}, I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Where j_x, j_y, j_z are moments of inertia of the spacerobot corresponding to x, y and z axis, respectively. Also ω_i is vector of angular velocity of pendulum masses. Then apply equations 1-3 with considering:

$$\tau_t = \begin{bmatrix} F \cos \delta_1 \cos \delta_2 \\ F \sin \delta_2 \\ -F \sin \delta_1 \cos \delta_2 \end{bmatrix},$$

$$\tau_r = \begin{bmatrix} M_x \\ M_y - F(b + d) \sin \delta_1 \cos \delta_2 \\ M_z - F(b + d) \sin \delta_2 \end{bmatrix}$$

So, the equations of the motion are derived:

$$(mb + m_0 h_0)(a_x + \Omega_2^2 + \Omega_3^2) + \sum_{i=1}^N m_i (a_x + l_i (\ddot{\varphi}_i \sin \varphi_i + \dot{\varphi}_i^2 \cos \varphi_i + 2 \dot{\varphi}_i \Omega_2 \cos \varphi_i \cos \theta_i - 2 \dot{\theta}_i \Omega_2 \sin \varphi_i \sin \theta_i - 2 \dot{\varphi}_i \Omega_3 \cos \varphi_i \sin \theta_i - 2 \dot{\theta}_i \Omega_3 \sin \varphi_i \cos \theta_i + \dot{\Omega}_2 \sin \varphi_i \cos \theta_i - \dot{\Omega}_3 \sin \varphi_i \sin \theta_i + (\Omega_2^2 - \Omega_3^2)(h_i - l_i \cos \varphi_i) + \Omega_1 \Omega_3 \sin \varphi_i \cos \theta_i + \Omega_1 \Omega_2 \sin \varphi_i \sin \theta_i)) = F \cos \delta_1 \cos \delta_2 \quad (6)$$

$$(m + m_0)a_y - (mb + m_0h_0)(\dot{\Omega}_3 + \Omega_1\Omega_2) + \sum_{i=1}^2 m_i (a_y + l_i (\ddot{\varphi}_i \cos \varphi_i \sin \theta_i + \ddot{\theta}_i \sin \varphi_i \cos \theta_i - \dot{\varphi}_i^2 \sin \varphi_i \sin \theta_i - \dot{\theta}_i^2 \sin \varphi_i \sin \theta_i + 2\dot{\varphi}_i \Omega_1 \cos \varphi_i \cos \theta_i + 2\dot{\theta}_i \Omega_1 \sin \varphi_i \sin \theta_i + 2\dot{\varphi}_i \Omega_3 \sin \varphi_i + 2\dot{\theta}_i \dot{\varphi}_i \cos \varphi_i \cos \theta_i - (\dot{\Omega}_1 - \Omega_2 \Omega_3) \sin \varphi_i \cos \theta_i - (\Omega_1^2 + \Omega_3^2) \sin \varphi_i \sin \theta_i + (\dot{\Omega}_3 + \Omega_1 \Omega_2)(h_i - l_i \cos \varphi_i))) = F \sin \delta_2 \quad (7)$$

$$(m + m_0)a_z + (mb + m_0h_0)(\dot{\Omega}_2 - \Omega_1\Omega_3) + \sum_{i=1}^2 m_i (a_z + l_i (\ddot{\varphi}_i \cos \varphi_i \cos \theta_i - \ddot{\theta}_i \sin \varphi_i \sin \theta_i - \dot{\varphi}_i^2 \sin \varphi_i \cos \theta_i - \dot{\theta}_i^2 \sin \varphi_i \cos \theta_i + 2\dot{\varphi}_i \Omega_1 \cos \varphi_i \sin \theta_i + 2\dot{\theta}_i \Omega_1 \sin \varphi_i \cos \theta_i - 2\dot{\varphi}_i \Omega_2 \sin \varphi_i - 2\dot{\theta}_i \dot{\varphi}_i \cos \varphi_i \sin \theta_i + (\dot{\Omega}_1 + \Omega_2 \Omega_3) \sin \varphi_i \sin \theta_i - (\Omega_1^2 + \Omega_2^2) \sin \varphi_i \sin \theta_i - (\dot{\Omega}_2 - \Omega_1 \Omega_3)(h_i - l_i \cos \varphi_i))) = -F \sin \delta_1 \cos \delta_2 \quad (8)$$

$$\dot{\Omega}_1(J_x + I_0) + \Omega_2 \Omega_3(J_z - J_y) + \sum_{i=1}^2 m_i (l_i (\sin \varphi_i \cos \theta_i (-\dot{\Omega}_3(h_i - l_i \cos \varphi_i) + 2\dot{\theta}_i \Omega_2(h_i - l_i \cos \varphi_i) - a_y - \Omega_1 \Omega_2(h_i - l_i \cos \varphi_i)) + \sin \varphi_i \sin \theta_i (-\dot{\Omega}_2(h_i - l_i \cos \varphi_i) + 2\dot{\theta}_i \Omega_3(h_i - l_i \cos \varphi_i) + a_z) + \dot{\varphi}_i \cos \varphi_i \sin \theta_i (2\Omega_2(h_i - l_i \cos \varphi_i))) + I_{x_i}(\ddot{\varphi}_i + \dot{\Omega}_1) + I_{z_i}(\ddot{\theta}_i \Omega_2 + \Omega_2 \Omega_3) - \Omega_3 v_z(h_i - l_i \cos \varphi_i))) = M_x \quad (9)$$

$$(mb + m_0h_0)(\dot{\Omega}_2 + a_z) + \dot{\Omega}_2(J_y + I_0) + \Omega_1 \Omega_3(J_x - J_z) + \sum_{i=1}^2 m_i (l_i (\sin \varphi_i \cos \theta_i (-\Omega_3^2(h_i - l_i \cos \varphi_i) + \dot{v}_x - \Omega_3 v_y) + \sin \varphi_i \sin \theta_i (2\Omega_3 v_z - 2\Omega_1 v_x) - (h_i - l_i \cos \varphi_i)(-2\dot{\varphi}_i \Omega_2 \sin \varphi_i + \Omega_2 \Omega_3 \sin \varphi_i \sin \theta_i)) - (h_i - l_i \cos \varphi_i)((-\dot{\Omega}_2 + \Omega_1 \Omega_3)(h_i - l_i \cos \varphi_i) + a_z) + I_{y_i} \dot{\Omega}_2 + I_{x_i} \Omega_3(\dot{\varphi}_i + \Omega_1) - I_{z_i} \Omega_1(\dot{\theta}_i + \Omega_3)) = M_y - F(b + d) \sin \delta_1 \cos \delta_2 \quad (10)$$

$$(mb + m_0h_0)(\dot{\Omega}_3 - a_y) + \dot{\Omega}_3(J_z + I_0) + \Omega_1 \Omega_2(J_y - J_x) + \sum_{i=1}^2 m_i (l_i (\sin \varphi_i \sin \theta_i (a_x + \Omega_2^2(h_i - l_i \cos \varphi_i)) + \sin \varphi_i \cos \theta_i (\Omega_2 \Omega_3(h_i - l_i \cos \varphi_i) - 2\Omega_2 v_y) - \Omega_1 \Omega_2 \cos \varphi_i) + (h_i - l_i \cos \varphi_i)(a_y + (\dot{\Omega}_3 + \Omega_1 \Omega_2)(h_i - l_i \cos \varphi_i)) + I_{z_i}(\ddot{\theta}_i + \dot{\Omega}_3) + I_{x_i} \Omega_2(-\dot{\varphi}_i - \Omega_1) + I_{y_i} \Omega_1 \Omega_2) = M_z - F(b + d) \sin \delta_2 \quad (11)$$

$$\sum_{i=1}^2 m_i (l_i (\sin \varphi_i (\dot{v}_x + l_i \ddot{\varphi}_i \sin \varphi_i + l_i \dot{\varphi}_i^2 \cos \varphi_i) + \cos \varphi_i \sin \theta_i (\dot{\Omega}_3(h_i - l_i \cos \varphi_i) + \dot{v}_y) + \dot{\varphi}_i \sin \varphi_i \sin \theta_i (\Omega_3(h_i - l_i \cos \varphi_i) + v_y) + \dot{\theta}_i \cos \varphi_i \cos \theta_i (\Omega_3(h_i - l_i \cos \varphi_i) + v_y) + \cos \varphi_i \cos \theta_i (-\dot{\Omega}_2(h_i - l_i \cos \varphi_i) + \dot{v}_z) + \dot{\varphi}_i \sin \varphi_i \cos \theta_i (\Omega_2(h_i - l_i \cos \varphi_i)) + \dot{\theta}_i \sin \varphi_i \cos \theta_i (\Omega_2(h_i - l_i \cos \varphi_i)) - (\Omega_3^2 + \Omega_2^2) \sin \varphi_i (h_i - l_i \cos \varphi_i) + v_x(-\Omega_3 \sin \varphi_i + \Omega_1 \cos \varphi_i \cos \theta_i + \dot{\varphi}_i \sin \varphi_i \sin \theta_i - \dot{\theta}_i \cos \varphi_i \cos \theta_i) + v_z(\Omega_2 \sin \varphi_i - \Omega_1 \cos \varphi_i \cos \theta_i))) + I_{x_i}(\ddot{\varphi}_i + \dot{\Omega}_1) + \epsilon_i \dot{\varphi}_i = 0 \quad (12)$$

$$\sum_{i=1}^2 m_i (l_i (\sin \varphi_i \cos \theta_i (\dot{\Omega}_3(h_i - l_i \cos \varphi_i) + \dot{v}_y) + \dot{\varphi}_i \cos \varphi_i \cos \theta_i (\Omega_3(h_i - l_i \cos \varphi_i)) + \dot{\theta}_i \sin \varphi_i \sin \theta_i (-\Omega_3(h_i - l_i \cos \varphi_i)) + \sin \varphi_i \sin \theta_i (\dot{\Omega}_2(h_i - l_i \cos \varphi_i) - \dot{v}_z) + \dot{\varphi}_i \cos \varphi_i \sin \theta_i (\Omega_2(h_i - l_i \cos \varphi_i)) + \dot{\theta}_i \sin \varphi_i \cos \theta_i (\Omega_2(h_i - l_i \cos \varphi_i)) + v_x(\Omega_3 \sin \varphi_i \cos \theta_i + \Omega_2 \sin \varphi_i \sin \theta_i) - v_y \Omega_1 \sin \varphi_i \sin \theta_i - v_z \Omega_1 \sin \varphi_i \cos \theta_i)) + I_{z_i}(\ddot{\theta}_i + \dot{\Omega}_3)) + \epsilon_i \dot{\theta}_i = 0 \quad (13)$$

Where I_{x_i} , I_{y_i} , I_{z_i} are moments of inertia of i-th mass corresponding to x, y and z-axis, respectively, and $a = [a_x \ a_y \ a_z]^T = \ddot{V} + \Omega \times V$ is the acceleration of the center of mass of the fuel in spacerobot frame.

Remark: we ignore terms that contain more than two trigonometric functions to simplify equations of motion. But in simulating all terms are considered.

III. CONTROLLER DESIGN

In this section, we design a feedback controller based on candidate Lyapunov function for spacerobot and fuel slosh dynamics system that it's formulated in previous section. It is don't necessary to control the velocity vector along the spacecraft's longitudinal axis, so for simplicity ignore the first equation of the equations of motion. Now by changing the old control inputs (torques about rigidly body and gimbal deflection angles) to new control inputs (u_{1-5}), equations of motions can be written as reduced form:

$$\dot{v}_y = u_1 - v_x \Omega_3 + v_z \Omega_1 \quad (14)$$

$$\dot{v}_z = u_2 - v_y \Omega_1 + v_x \Omega_2 \quad (15)$$

$$\dot{\Omega}_1 = u_3 \quad (16)$$

$$\dot{\Omega}_2 = u_4 \quad (17)$$

$$\dot{\Omega}_3 = u_5 \quad (18)$$

$$\ddot{\varphi}_i = \left(\frac{-1}{(l_i^2 \sin^2 \varphi_i + I_{x_i})} \right) (l_i (\sin \varphi_i (\dot{v}_x + l_i \dot{\varphi}_i^2 \cos \varphi_i) + \cos \varphi_i \sin \theta_i (\Omega_3 (h_i - l_i \cos \varphi_i) + \dot{v}_y) + \dot{\varphi}_i \sin \varphi_i \sin \theta_i (\Omega_3 (h_i - l_i \cos \varphi_i) + v_y) + \dot{\theta}_i \cos \varphi_i \cos \theta_i (\Omega_3 (h_i - l_i \cos \varphi_i) + v_y) + \cos \varphi_i \cos \theta_i (-\Omega_2 (h_i - l_i \cos \varphi_i) + \dot{v}_z) + \dot{\varphi}_i \sin \varphi_i \cos \theta_i (\Omega_2 (h_i - l_i \cos \varphi_i)) + \dot{\theta}_i \sin \varphi_i \cos \theta_i (\Omega_2 (h_i - l_i \cos \varphi_i)) - (\Omega_3^2 + \Omega_2^2) \sin \varphi_i (h_i - l_i \cos \varphi_i) + v_x (-\Omega_3 \sin \varphi_i + \Omega_1 \cos \varphi_i \cos \theta_i + \dot{\varphi}_i \sin \varphi_i \sin \theta_i - \dot{\theta}_i \cos \varphi_i \cos \theta_i) + v_z (\Omega_2 \sin \varphi_i - \Omega_1 \cos \varphi_i \cos \theta_i)) + I_{x_i} \dot{\Omega}_1 - \frac{\varepsilon_i \dot{\varphi}_i}{m_i}) \quad (19)$$

$$\ddot{\theta}_i = \left(\frac{-1}{I_{z_i}} \right) (l_i (\sin \varphi_i \cos \theta_i (\Omega_3 (h_i - l_i \cos \varphi_i) + \dot{v}_y) + \dot{\varphi}_i \cos \varphi_i \cos \theta_i (\Omega_3 (h_i - l_i \cos \varphi_i)) + \dot{\theta}_i \sin \varphi_i \sin \theta_i (-\Omega_3 (h_i - l_i \cos \varphi_i)) + \sin \varphi_i \sin \theta_i (\Omega_2 (h_i - l_i \cos \varphi_i) - \dot{v}_z) + \dot{\varphi}_i \cos \varphi_i \sin \theta_i (\Omega_2 (h_i - l_i \cos \varphi_i)) + \dot{\theta}_i \sin \varphi_i \cos \theta_i (\Omega_2 (h_i - l_i \cos \varphi_i)) + v_x (\Omega_3 \sin \varphi_i \cos \theta_i + \Omega_2 \sin \varphi_i \sin \theta_i) - v_y \Omega_1 \sin \varphi_i \sin \theta_i - v_z \Omega_1 \sin \varphi_i \cos \theta_i) + I_{z_i} \dot{\Omega}_3 - \frac{\varepsilon_i \dot{\theta}_i}{m_i}) \quad (20)$$

With considering the reduced equations of pendulums motion ($\dot{\varphi}, \dot{\theta}$), we present the candidate Lyapunov function:

$$\mathbf{V} = \frac{1}{2} (r_1 v_y^2 + r_2 v_z^2 + r_3 \vartheta_1^2 + r_4 \Omega_1^2 + r_5 \vartheta_2^2 + r_6 \Omega_2^2 + r_7 \vartheta_3^2 + r_8 \Omega_3^2) + \frac{1}{2} \sum_{i=1}^2 m_i (r_9 (\dot{\varphi}_i^2 + \left(\frac{-1}{(l_i^2 \sin^2 \varphi_i + I_{x_i})} \right) (-2 \dot{\varphi}_i \Omega_3 h_i l_i \cos \varphi_i \sin \theta_i + 2 \dot{\varphi}_i \Omega_2 h_i l_i \cos \varphi_i \cos \theta_i - 2 \dot{\varphi}_i \Omega_1 I_{x_i}) + r_{10} (\dot{\theta}_i^2 + \left(\frac{-1}{I_{z_i}} \right) (-2 \dot{\theta}_i \Omega_3 I_{z_i} - 2 \dot{\theta}_i \Omega_3 h_i l_i \sin \varphi_i \cos \theta_i - 2 \dot{\theta}_i \Omega_2 h_i l_i \sin \varphi_i \sin \theta_i)) \quad (21)$$

Where $\vartheta = [\vartheta_1 \ \vartheta_2 \ \vartheta_3]^T$ is vector of attitude angles of the spacerobot, and r_{1-10} are arbitrary values. For stability, the candidate Lyapunov function has to be a positive definition on domain of pendulum's motion, and also time derivative of candidate Lyapunov function has to be a negative definition on this domain. New controller can be achieved by using the time derivative of the candidate Lyapunov function, that it's a function as:

$$\dot{\mathbf{V}} = [A] + [B]u_1 + [C]u_2 + \sum_{i=1}^3 (f_{i1}u_3 + f_{i2}u_4 + f_{i3}u_5 + g_i)\Omega_i \quad (22)$$

Multiplication of [B] and correction coefficient (w_1) produce u_1 , and multiplication of [C] and correction coefficient (w_2) produce u_2 . The other controllers (u_{3-5}) can be produce as:

$$\begin{bmatrix} u_3 \\ u_4 \\ u_5 \end{bmatrix} = - \begin{bmatrix} f_{11} & f_{12} & f_{13} \\ f_{21} & f_{22} & f_{23} \\ f_{31} & f_{32} & f_{33} \end{bmatrix}^{-1} \begin{bmatrix} g_1 + w_3 \Omega_1 \\ g_2 + w_4 \Omega_2 \\ g_3 + w_5 \Omega_3 \end{bmatrix} \quad (23)$$

Where w_{1-5} are arbitrary and positive values.

IV. SIMULATION

In this section for describe effectiveness of nonlinear feedback control law that it's designed in previous section, a physical example is simulated. Here, a two pendulums model is considered for characterization two sloshing modes of fuel. The control objectives are stabilization attitude angles and transverse velocity of the spacerobot in orbital maneuvering. The control gains are given in Table. 1, and the physical parameters are given in Table. 2.

TABLE I. CONTROL GAINS

$r_4 = r_6 = r_8 = 500$	$r_1 = r_2 = 8 \times 10^{-7}$
$r_9 = r_{10} = 1 \times 10^{-5}$	$r_3 = r_5 = r_7 = 1 \times 10^3$
$w_3 = w_4 = w_5 = 4 \times 10^4$	$w_1 = w_2 = 1 \times 10^4$

TABLE II. PHYSICAL PARAMETERS

$m = 975$	Kg	$h_0 = 0.05$	m
$m_0 = 480$		$h_1 = 0.6$	
$m_1 = 50$		$h_2 = 0.9$	
$m_2 = 5$		$b = -0.6$	
$I_{x_1} = I_{y_1} = I_{z_1} = 10$	Kg.m ²	$d = 1.2$	N
$I_{x_2} = I_{y_2} = I_{z_2} = 1$		$F = 2450$	
$l_1 = 0.2$	m	$\varepsilon_1 = 3.7$	Kg.m ² /s
$l_2 = 0.1$		$\varepsilon_2 = 0.5$	

Then, simulation is done by considering the reduced equations of motion (14-20), applying the controllers that they are designed by candidate Lyapunov function, and put the parameters and gains in these equations.

As seen in Fig.3 and Fig.4 axial transverse velocity and attitude angles of the spacecraft after elapse 700 second asymptotically tend to zero, when, the value of axial longitudinal velocity increases, also as seen in Fig. 5 and Fig. 6 the position angle of pendulums about x-axis (φ) asymptotically tend to zero, and the position angle of pendulums about z-axis (θ) asymptotically becomes stable.

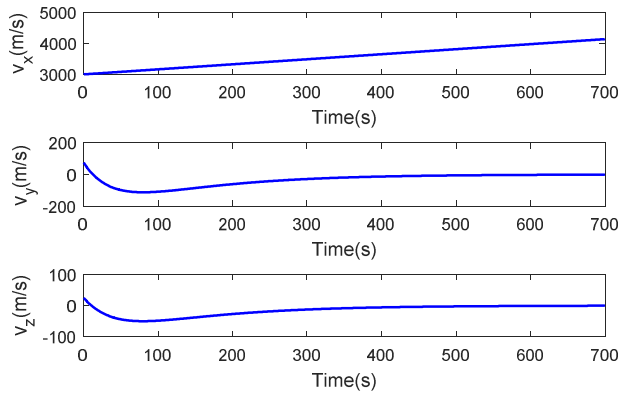


Fig. 3. Time responses of v_x , v_y and v_z

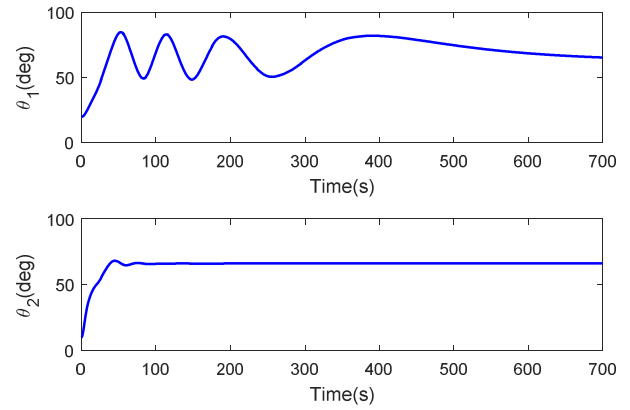


Fig. 6. Time responses of θ_1 and θ_2

V. CONCLUSION

This paper presents a 3D multi-pendulum model for characterization the most prominent sloshing modes of fuel of the spacerobots that it has a fuel container and maneuver in 3D space, then a nonlinear feedback control based on candidate Lyapunov function is designed to control fuel sloshing, and at the end, effectiveness of designed controller is described by a physical example simulation. As seen in the results of simulation, the designed controller has been successful to control fuel sloshing.

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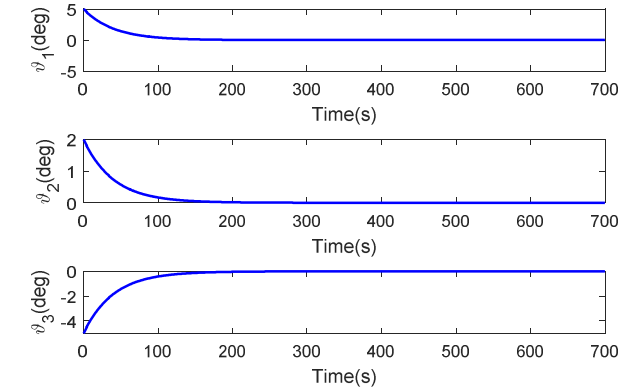


Fig. 4. Time responses of φ_1 , φ_2 and φ_3

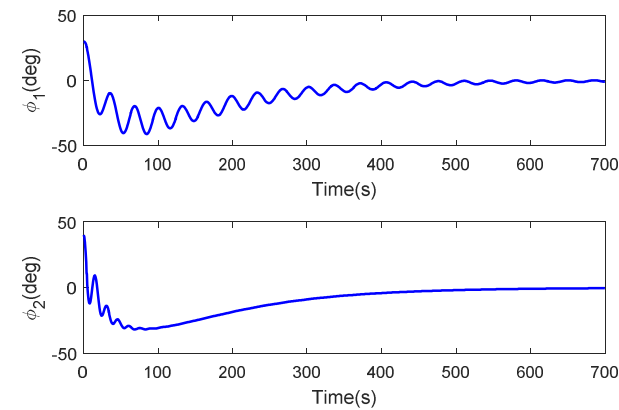


Fig. 5. Time responses of φ_1 and φ_2

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